

Zongrong (Jasper) Li

Department of Geography, College of Arts & Sciences, Texas A&M University

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Education

Texas A&M University — Ph.D. in Geography College Station, TX
Advisor: Prof. Lei Zou (GEAR Lab); GPA 4.0/4.0; Dean's Doctoral Fellowship (\$40,000) Aug. 2025 – May 2029 (expected)

University of Southern California — M.S. in Spatial Data Science Los Angeles, CA
(Rank 1%)
Advisors: Prof. Siqin Wang & Prof. Yi Qi (Spatial Sciences Institute); GPA 3.92/4.0 Aug. 2023 – May 2025

Nanjing Audit University — B.Econ. in Finance (Rank 10%) Nanjing, China
GPA 3.53/5.0 Sep. 2018 – Jun. 2022

Research Interests

Geospatial Artificial Intelligence (GeoAI) & Geospatial Foundation Models • Generative Digital Twins • 3D Urban Modeling • Built (Indoor) Environment & Human–Environment Interaction • Geographic Information Science (GIScience)

Publications

Li, Z. is the author. *Corresponding author; co-first authorship as noted. IF / JCR quartile shown where applicable. Citation metrics (Google Scholar, Aug. 2026): 56 citations, h-index 4, i10-index 2.

- [1] **Li, Z.**, Yang, M., Zou, L.* , Ma, H., Tian, H., Zhou, S., Gong, W., Zhang, K., Chen, B., Zhang, M., & Yang, Y. (2026). GIScholarBench: Benchmarking LLM Overconfidence in GIS Research. *Submitted to ACM SIGSPATIAL 2026*. arXiv:2606.08036
- [2] **Li, Z.**, Zou, L.* , Seo, J., Yang, M., & Yang, Y. (2026). Seeing Green from Indoors in 3D: How Built Environment and Vegetation Shape Window-Level Nature Exposure. *Landscape and Urban Planning* (Under Review). (IF 9.2, Q1)
- [3] **Li, Z.**, Li, H., Yang, Y., Wang, S.* , & Zhu, Y. (2026). Integrating Earth Observation Data into the Tri-environmental Evaluation of the Economic Cost of Natural Disasters: A Case Study of the 2025 LA Wildfire. *Int. J. Applied Earth Observation and Geoinformation*, 150, 105342. doi:10.1016/j.jag.2026.105342 (IF 8.6, Q1)
- [4] **Li, Z.**, Zhang, F., Dai, S., & Zhao, W.* (2026). Bridging Street View Coverage Disparities through Geographic Style-Preserving Generation from Satellite View. *ISPRS Journal of Photogrammetry and Remote Sensing*, 236, 622–639. doi:10.1016/j.isprsjprs.2026.03.049 (IF 12.2, Q1)
- [5] **Li, Z.**, Su, Y., Biljecki, F., & Zhao, W.* (2026). BuildingMultiView: Powering Multi-scale Building Characterization with Large Language Models and Multi-perspective Imagery. *Int. J. Applied Earth Observation and Geoinformation*, 146, 105034. doi:10.1016/j.jag.2025.105034 (IF 8.6, Q1)
- [6] Yang, M., Zou, L.* , Ma, H., **Li, Z.**, & Li, W. (2025). The Geography of Insufficient Sleep in the Contiguous United States (CONUS). EarthArXiv:10810

- [7] Wang, S.*, Wang, Z., Huang, X., Liu, L., Shi, Z., & **Li, Z.** (2025). Benchmark Evaluation of Large Language Models for Human versus AI Perception on Urban Environment through Street View Images. *Landscape and Urban Planning* (Under Review). (IF 9.2, Q1)
- [8] **Li, Z.**, Xu, J., Wang, S.*, Wu, Y., & Li, H. (2025). StreetViewLLM: Extracting Geospatial Information Using Multimodal Chain-of-Thought Large Language Models. *Int. J. Applied Earth Observation and Geoinformation* (Under Second Round of Review). arXiv:2411.14476 (IF 8.6, Q1)
- [9] **Li, Z.**, Wang, J., Xu, J., Li, B., & Wang, S.* (2025). ViT-Based Attention Mechanism Uncovers Disparities in Urban Economic Landscapes Across Cities. *In Preparation*.
- [10] **Li, Z.**, Su, Y., Zhu, C., & Zhao, W.* (2024). BuildingView: Constructing Urban Building Exteriors Databases with Street View Imagery and Multimodal Large Language Model. *Accepted to ACM SpatialDI 2025*. arXiv:2409.19527
- [11] **Li, Z.**, Li, Q., Li, H., Wang, S.*, & Qi, Y.* (2024). Multi-source Tri-environmental Conceptual Framework for Fire Impact Analysis. *Urban Informatics*, 3, 31. doi:10.1007/s44212-024-00063-7
- [12] Yang, Y., Zou, L.*, Gong, W., Fu, K., **Li, Z.**, Wang, S., Zhou, B., Cai, H., & Tian, H. (2026). DamageArbiter: A CLIP-Enhanced Multimodal Arbitration Framework for Hurricane Damage Assessment from Street-View Imagery. arXiv:2603.14837
- [13] Yang, Y., Gong, W., Zhang, K., Zou, L., Tu, Z., Li, H., **Li, Z.**, & Ye, X. (2026). RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. *Accepted to ACM SIGSPATIAL 2026*. arXiv:2606.21819

Research Experience

Graduate Research Assistant, TAMU GEAR Lab & Google

Mentors: Prof. Lei Zou & Dr. Hongxu Ma

Aug. 2025 – Present

- Develop **GIScholarBench**, a large-scale LLM benchmark for GIScience literature retrieval and knowledge discovery, curating 17,000+ papers from 25 core journals (2020–2025) via CrossRef, OpenAlex, and Semantic Scholar APIs.
- Evaluate retrieval, citation-linking, and hypothesis-generation across frontier LLMs (OpenAI, Gemini, xAI) to analyze human–AI research behavior, reasoning bias, and novelty-driven knowledge discovery. Funded by TAMIDS GAIRE Grant (2026–2027, \$7,000).

Research Assistant, HKUST (Guangzhou) — *Mentor: Prof. Wufan Zhao* *Jun. 2024 – Sep. 2025*

- Developed **BuildingView**, integrating OpenStreetMap, Google Street View, and remote sensing with multimodal LLMs to build urban building-exterior databases for New York, Amsterdam, and Singapore; validated against third-party real-estate transaction data.

Research Assistant, USC Spatial Sciences Institute

Mentors: Prof. Siqin Wang, Prof. Yi Qi

Sep. 2023 – May 2025

- Built a multi-source tri-environmental framework (optical, thermal, demographic) for 2023 Hawaii wildfire damage estimation, and **StreetViewLLM** for multimodal Chain-of-Thought geospatial extraction.

Research Assistant, Harvard University — *Spatial Data Lab, Center for Geographic Analysis* *May 2024 – Sep. 2024*

- TODO: one line on what you actually did at the Spatial Data Lab.

Research Assistant, The Hong Kong Polytechnic University — *Mentor: Prof. Zhiwei Li* *Sep. 2023 – Aug. 2024*

- Assessed global flood impacts on megacities using Google Earth Engine, VIIRS nighttime-light data, and global flood databases; analyzed post-flood recovery and regional vulnerability disparities.

Research Assistant, Nanjing Audit University — *Mentor: Prof. Chong Peng* Sep. 2019 – May 2021

- Analyzed 3+ TB of urban data (traffic, housing density, amenities); mapped transportation inefficiencies with GIS and supported a municipal geospatial data warehouse.

Industry Experience

Founding GTM/Scientist, Monid, Inc.

Supervisor: Shengkun Ye (Chief Executive Officer)

Aug. 2026 – Present

- Build geodemographic segmentation of the customer base from spatial and firmographic data; analyze the geographic and firmographic distribution of user acquisition to identify high-yield market segments.
- Quantify acquisition-channel performance variation across regions and market types using spatial statistics; design and evaluate cohort and A/B experiments on conversion, lift, and retention.
- Maintain reproducible pipelines for market and customer data retrieval and validation.

Data Science Intern, City of Hope — *Mentor: Dr. Jay Yang*

Jan. 2024 – Sep. 2024

- Evaluated deep learning models (CNNs, GNNs) on satellite imagery for urban-health assessment, studying urban form, structural racism, and environmental risk (air pollution, urban heat islands); validated with demographic data.

Data Analyst Intern, NetEase Youdao — *Business Analysis Department*

Jan. 2023 – Apr. 2023

- Cleaned and standardized sales data across 50+ regions and 2,000+ colleges; built a Tableau data map identifying a potential 30% sales-conversion lift from localized strategy; delivered an online-education industry report.

Audit Intern, BDO — *Energy Company Group*

Dec. 2021 – Jan. 2022

- Supported the annual audit of an energy company and 7 subsidiaries; audited fixed assets, intangibles, profits, and receivables; prepared 25 confirmation letters and 25 working papers.

Teaching Experience

Teaching Assistant, Texas A&M University

*GEOG-391/659 Geo-Database; GEOG-392/500 GIS Programming;
GEOG-475 Advanced GIS*

Aug. 2025 – Present

College Station, TX

Graduate Grader, University of Southern California

*SSCI-581 Concepts for Spatial Thinking; SSCI-583 Spatial Analysis
and Modeling*

Jan. 2024 – May 2025

Los Angeles, CA

Honors & Awards

Doctoral Student Research, Travel & Professional Development Award (\$1,000)	Apr. 2026
AAG Cartography and Mapping Specialty Group Session Funding Award (\$202)	Apr. 2026
Google Cloud Research Credits (\$1,000)	Feb. 2026
AAG International Geographic Information Fund (\$1,200)	Feb. 2026
AAG AGSG Annual Meeting Award (\$202)	Feb. 2026
TAMU Lechner Graduate Scholarship (\$1,000)	Aug. 2025
TAMU Dean's Doctoral Fellowship (\$40,000)	Aug. 2025
Lifetime Membership — Nu Theta Chapter, Gamma Theta Upsilon	May 2025
AAG GISS Student Paper Competition — Honorable Mention (\$300)	Mar. 2025
EIP Student of the Year Award (\$500)	Mar. 2025
LA GeoSpatial Summit — Most Suitably Applied Analysis Methodology Award	Feb. 2025
Fall GSG Professional Development Fund (\$500)	Feb. 2025
Finalist, GISS Student Honors Paper Competition	Dec. 2024
LA GeoSpatial Summit — Story Map Participation Award	Feb. 2024
Bronze Medal, Kaggle Child Mind Institute — Detect Sleep States	Jan. 2023
“Outstanding Graduate” (Top 5%), Nanjing Audit University	May 2022
Provincial Champion, The World Mandarin Debating Championship	Sep. 2020

Selected Presentations

- **2026 ACSP 66th Annual Conference** (Pittsburgh, Oct. 2026) — Accepted individual paper presentation (upcoming).
- **2026 CaGIS/UCGIS Conference** (St. Louis, Sep. 2026) — Registered participant (upcoming).
- **2026 AAG Annual Meeting** — Organizer (GISER Symposia: Human–Environment Interactions in Digital Twin and 3D Urban Space; Past, Present, and Future of GIScience (Panel); Less Is More? The Value of Small Geospatial Data in the Era of Big Data and AI) & Presenter (“Seeing Green from Indoors in 3D”).
- **2025 TAMU GIS Day** — “Bridging Street View Coverage Disparities through Geographic Style-Preserving Generation.”
- **2025 AAG Annual Meeting** — “StreetViewLLM: Extracting Geospatial Information Using Multimodal Chain-of-Thought LLMs.”
- **2025 LA GeoSpatial Summit** — “Evaluating Los Angeles’ Transit Equity for the 2028 Car-Free Olympics.”
- **2025 Young Scholar Symposium on Spatiotemporal Data Science** — “Multi-source Tri-environmental Conceptual Framework for Fire Impact Analysis.”
- **2024 LA GeoSpatial Summit** — “Paradise on Fire: Cost-Effective Rapid 2023 Hawaii Wildfire Analysis.”
- **2024 Esri User Conference & Esri Fall Education Open House** — Participant.

Academic Service

Student Ambassador: TAMU Institute of Data Science (TAMIDS) — Generative AI Student Ambassador, 2026–2027 cohort (selected Jul. 2026).

Student Representative: TAMU AGGS Executive Committee (2026–2027); AAG Applied Geography Specialty Group (2026–2029); AAG CyberInfrastructure Specialty Group (2026–2027).

Founder: AutonomousGeoAI4Science (open AI & geospatial research community); AutoApplyClaw (AI

job-application platform for international students).

Science Communication: Study-abroad and research-life creator on RedNote (Xiaohongshu), 30,000+ followers.

Journal Reviewer (34 review reports across 13 journals, including 4 Nature-portfolio):

- Computational Urban Science (13); Scientific Reports (4); Humanities & Social Sciences Communications (3); npj Heritage Science (2); npj Natural Hazards (1)
- Photogrammetric Engineering & Remote Sensing (2); Applied Spatial Analysis and Policy (2); Environmental Research: Health (2); International Journal of Digital Earth (1); Journal of Computing in Civil Engineering (1); Discover Applied Sciences (1); Frontiers in Forests and Global Change (1); Forum Geographi (1)

Selected Projects & Open-Source Repositories

- **AutoApplyClaw** — AI job-application platform: job search, tailored resumes and cover letters, one-click form autofill via a Chrome extension (React, FastAPI, Claude API). autoapplyclaw.com
- **OfferOpenClaw** — Email-triggered AI agent automating PhD faculty outreach: scrapes faculty pages, rates research alignment, and generates personalized cold emails from a CV. github.com/Jasper0122/Offeropenclaw
- **Auto-Eco** — Automated econometrics paper-writing system, from data exploration to full draft. jasper0122.github.io/auto-eco
- **AAG 125th Anniversary Globe** — Interactive 3D globe visualizing all 125 AAG Annual Meeting locations (1904–2029). aag-125-globe.vercel.app
- **DeepLabV3Plus for Satellite Image Segmentation** (OpenEarthMap); **Foursquare Open-Source Places** (POI data processing); **Dasymetric Mapper** (high-resolution population density layer generation).
- **Yelp ChatDB** — Natural-language restaurant search system over a Yelp database. github.com/Jasper0122/yelp-chatdb-demo

Technical Skills

Programming: Python, JavaScript, PyTorch | **GIS / RS:** ArcGIS Pro, Google Earth Engine |
Expertise: Large Vision–Language Model Agents, 3D Web-GIS Development